Box Plots and Comparing Groups

Name _____ Date _____ No calculator Version 1

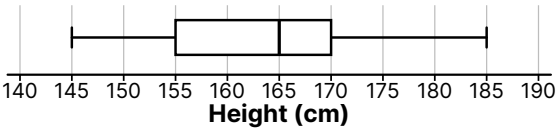
Scan to practise this topic online

KEY WORDS box plot *(kauwhata pouaka)* · quartile · whisker · skewed

A Reading a box plot ●○○

Remember: The box runs from the lower quartile to the upper quartile, with a line at the median. The whiskers reach the minimum and the maximum. The interquartile range (IQR) is the width of the box.

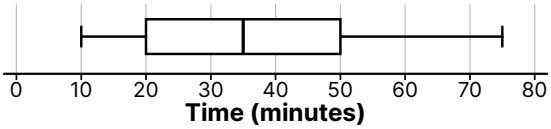
1. The box plot shows the heights, in centimetres, of the students in a class.



a) Minimum _____ **b)** Lower quartile _____ **c)** Median _____ **d)** Upper quartile _____

e) Maximum _____ **f)** Range _____ **g)** IQR _____

2. The box plot shows the times, in minutes, that a group of students spent on homework one night.



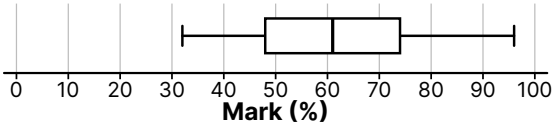
a) Minimum _____ **b)** Lower quartile _____ **c)** Median _____ **d)** Upper quartile _____

e) Maximum _____ **f)** Range _____ **g)** IQR _____

B The quarters of a box plot ●●○

Remember: Each piece of a box plot holds a quarter of the values: whisker, half box, half box, whisker. So 75% of the values are below the upper quartile and 75% are above the lower quartile.

3. The box plot shows the marks, out of 100, in a Year 10 test. There are 28 values in the data. What percentage of the values are:



a) below the median, 61%? _____ **b)** below the upper quartile, 74%? _____

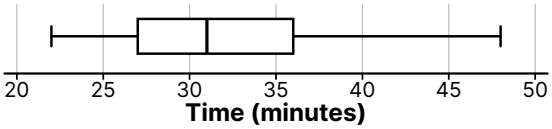
c) above the lower quartile, 48%? _____ **d)** inside the box, from 48% to 74%? _____

e) How many values are above the median? _____ **f)** How many are in the top quarter? _____

g) How many are in the lower 75%? _____ **h)** How many lie inside the box? _____

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4. The box plot shows how many minutes a group of people took to run 5 km. There are 40 values in the data. What percentage of the values are:



a) below the median, 31 min? _____ **b)** below the upper quartile, 36 min? _____

c) above the lower quartile, 27 min? _____ **d)** inside the box, from 27 min to 36 min? _____

e) How many values are above the median? ____ **f)** How many are in the top quarter? _____

g) How many are in the lower 75%? _____ **h)** How many lie inside the box? _____

C The shape of the data ●●○

Remember: Symmetric: the median sits in the middle of the box and the whiskers are about the same length. Skewed: the box and whisker are longer on one side, and the data stretch out that way.

5. Describe the shape of the data shown by each box plot: symmetric, skewed to the right or skewed to the left. Say what you can see in the box and the whiskers.

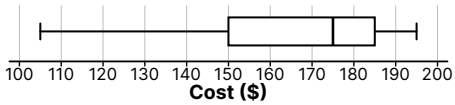
a)



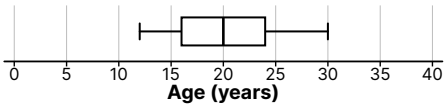
b)



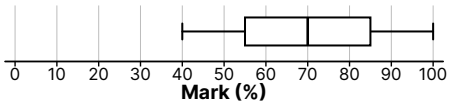
c)



d)



e)



f)

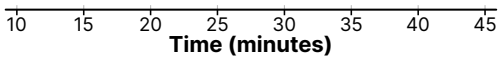


D Drawing a box plot ●●○

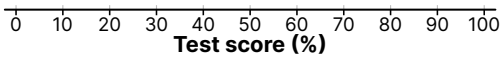
Remember: Mark the five numbers above the axis. Draw the box from LQ to UQ, a line at the median, and whiskers out to the minimum and maximum.

6. Draw a box plot above each axis.

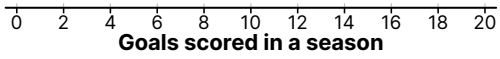
a) Minimum 12, LQ 18, median 24, UQ 30, maximum 42



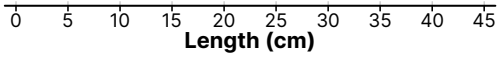
b) Minimum 25, LQ 45, median 60, UQ 70, maximum 95



c) Minimum 1, LQ 4, median 7, UQ 12, maximum 19



d) Data: 8, 12, 12, 15, 20, 22, 26, 30, 34, 36, 40. Find the five-point summary first.

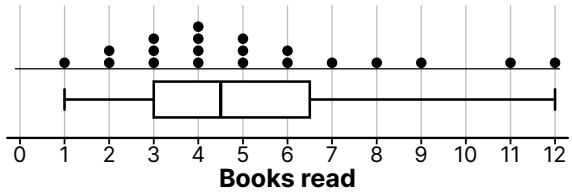


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E A dot plot and its box plot ●●○

Remember: A dot plot shows every value. The box plot summarises the same data: half the dots sit inside the box, a quarter on each whisker.

7. The dot plot shows the number of books read by a class in a term. The dot plot and the box plot show the same 20 values.



a) How many values are there? Count the dots. _____

b) What is the most common value? _____

c) Read the median from the box plot, then check it against the dots. _____

d) How many dots lie inside the box (from 3 to 6.5)? _____

e) How many dots are above the median? _____

f) Describe the shape of the data. _____

8. The dot plot shows the ages of the players in a club. The dot plot and the box plot show the same 20 values.



a) How many values are there? Count the dots. _____

b) What is the most common value? _____

c) Read the median from the box plot, then check it against the dots. _____

d) How many dots lie inside the box (from 13.5 to 16.5)? _____

e) How many dots are above the median? _____

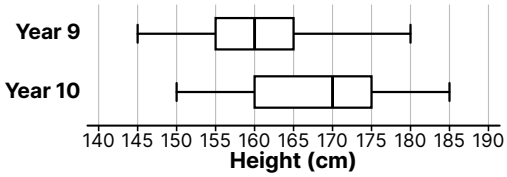
f) Describe the shape of the data. _____

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F Comparing two groups ●●●

Remember: Compare the middles (the medians), the spreads (the ranges and interquartile ranges) and the shapes.

9. The box plots show the heights of a group of Year 9 students and a group of Year 10 students.



a) Which group has the higher median? Write down both medians.

b) Find the range of each group.

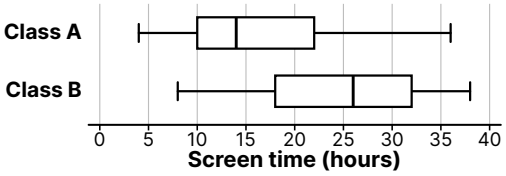
c) Find the interquartile range of each group.

d) In which group is the middle half of the data more spread out?

e) True or false: more than half of the Year 10 group are taller than 165 cm.

f) Describe the shape of each group.

10. The box plots show the weekly screen time, in hours, of two classes.



a) Which group has the higher median? Write down both medians.

b) Find the range of each group.

c) Find the interquartile range of each group.

d) In which group is the middle half of the data more spread out?

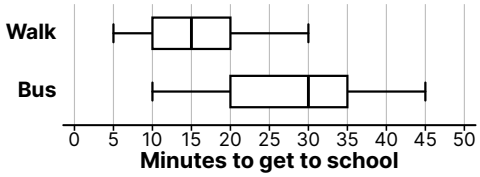
e) True or false: more than half of the Class B group spend more than 22 hours.

f) Describe the shape of each group.

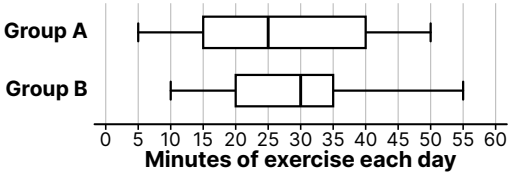
G Making the call ●●●

Remember: You can say one group tends to be higher when the median of one group is outside the box of the other. If each median sits inside the other box, the overlap is too big to make the call.

11. The box plots show how long students who walk and students who take the bus take to get to school. Do students who take the bus tend to take longer? Use the medians and the boxes to explain.

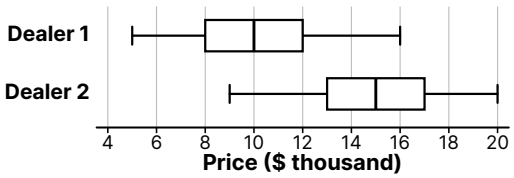


12. The box plots show the minutes of exercise each day for two groups of adults. Does Group B tend to exercise for longer? Use the medians and the boxes to explain.

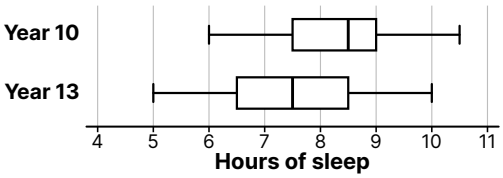


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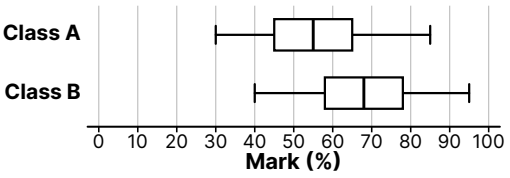
13. The box plots show the prices of second-hand cars at two dealers. Does Dealer 2 tend to charge more? Use the medians and the boxes to explain.



14. The box plots show the hours of sleep reported by Year 10 and Year 13 students. Do Year 10 students tend to sleep for longer? Use the medians and the boxes to explain.



15. The box plots show the marks of two classes in the same test. Does Class B tend to score higher? Use the medians and the boxes to explain.



Extension

16. A set of data has a minimum of 5 and a lower quartile of 14. Its interquartile range is 12, its median is 6 more than its lower quartile, and its range is 40.

a) Find the upper quartile. **b)** Find the median. **c)** Find the maximum.

17. A box plot of 60 values has minimum 12, LQ 20, median 26, UQ 34, maximum 50.

a) How many values are at least 20?

b) How many values are between 20 and 26?

c) How many values are above the upper quartile, 34?

d) How many values are below the median, 26?

e) Can you tell how many values are between 14 and 18? Explain.

f) Two sets of data have the same five-point summary. Must they have the same values? Explain.

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Box Plots and Comparing Groups

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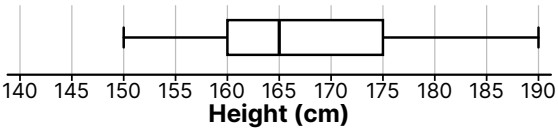
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KEY WORDS box plot *(kauwhata pouaka)* · quartile · whisker · skewed

A Reading a box plot ●○○

Remember: The box runs from the lower quartile to the upper quartile, with a line at the median. The whiskers reach the minimum and the maximum. The interquartile range (IQR) is the width of the box.

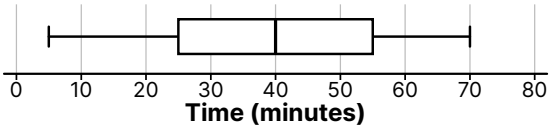
1. The box plot shows the heights, in centimetres, of the students in a class.



a) Minimum _____ **b)** Lower quartile _____ **c)** Median _____ **d)** Upper quartile _____

e) Maximum _____ **f)** Range _____ **g)** IQR _____

2. The box plot shows the times, in minutes, that a group of students spent on homework one night.



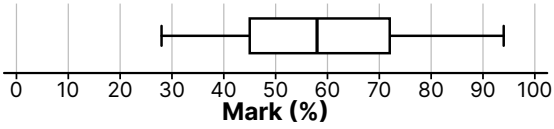
a) Minimum _____ **b)** Lower quartile _____ **c)** Median _____ **d)** Upper quartile _____

e) Maximum _____ **f)** Range _____ **g)** IQR _____

B The quarters of a box plot ●●○

Remember: Each piece of a box plot holds a quarter of the values: whisker, half box, half box, whisker. So 75% of the values are below the upper quartile and 75% are above the lower quartile.

3. The box plot shows the marks, out of 100, in a Year 10 test. There are 32 values in the data. What percentage of the values are:



a) below the median, 58%? _____ **b)** below the upper quartile, 72%? _____

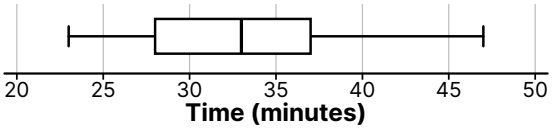
c) above the lower quartile, 45%? _____ **d)** inside the box, from 45% to 72%? _____

e) How many values are above the median? _____ **f)** How many are in the top quarter? _____

g) How many are in the lower 75%? _____ **h)** How many lie inside the box? _____

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4. The box plot shows how many minutes a group of people took to run 5 km. There are 48 values in the data. What percentage of the values are:



a) below the median, 33 min? _____ **b)** below the upper quartile, 37 min? _____

c) above the lower quartile, 28 min? _____ **d)** inside the box, from 28 min to 37 min? _____

e) How many values are above the median? ____ **f)** How many are in the top quarter? _____

g) How many are in the lower 75%? _____ **h)** How many lie inside the box? _____

C The shape of the data ●●○

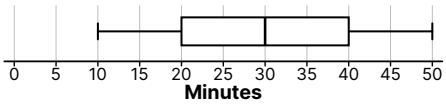
Remember: Symmetric: the median sits in the middle of the box and the whiskers are about the same length. Skewed: the box and whisker are longer on one side, and the data stretch out that way.

5. Describe the shape of the data shown by each box plot: symmetric, skewed to the right or skewed to the left. Say what you can see in the box and the whiskers.

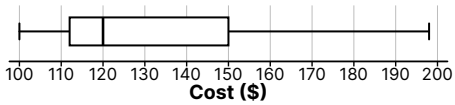
a)



b)



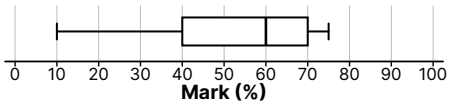
c)



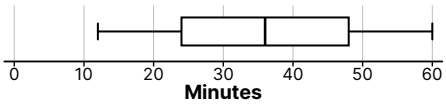
d)



e)



f)

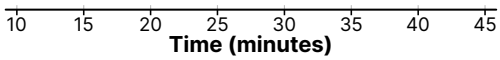


D Drawing a box plot ●●○

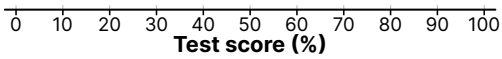
Remember: Mark the five numbers above the axis. Draw the box from LQ to UQ, a line at the median, and whiskers out to the minimum and maximum.

6. Draw a box plot above each axis.

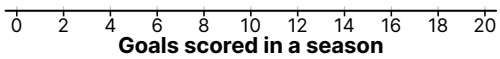
a) Minimum 14, LQ 20, median 26, UQ 34, maximum 44



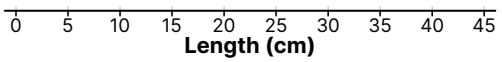
b) Minimum 15, LQ 40, median 55, UQ 75, maximum 90



c) Minimum 2, LQ 5, median 9, UQ 13, maximum 18



d) Data: 6, 10, 14, 16, 18, 24, 25, 28, 32, 38, 42. Find the five-point summary first.

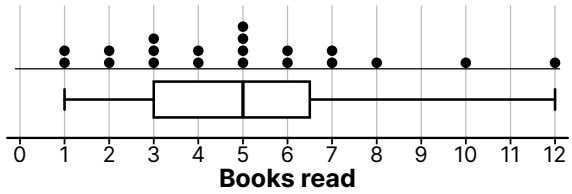


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E A dot plot and its box plot ●●○

Remember: A dot plot shows every value. The box plot summarises the same data: half the dots sit inside the box, a quarter on each whisker.

7. The dot plot shows the number of books read by a class in a term. The dot plot and the box plot show the same 20 values.



a) How many values are there? Count the dots. _____

b) What is the most common value? _____

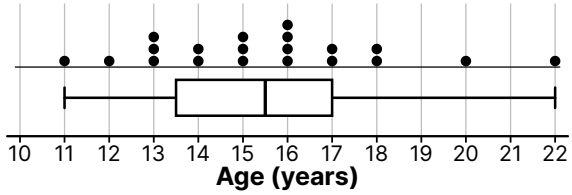
c) Read the median from the box plot, then check it against the dots. _____

d) How many dots lie inside the box (from 3 to 6.5)? _____

e) How many dots are above the median? _____

f) Describe the shape of the data. _____

8. The dot plot shows the ages of the players in a club. The dot plot and the box plot show the same 20 values.



a) How many values are there? Count the dots. _____

b) What is the most common value? _____

c) Read the median from the box plot, then check it against the dots. _____

d) How many dots lie inside the box (from 13.5 to 17)? _____

e) How many dots are above the median? _____

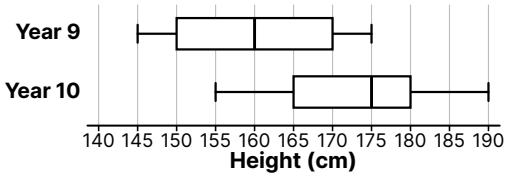
f) Describe the shape of the data. _____

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F Comparing two groups ●●●

Remember: Compare the middles (the medians), the spreads (the ranges and interquartile ranges) and the shapes.

9. The box plots show the heights of a group of Year 9 students and a group of Year 10 students.



a) Which group has the higher median? Write down both medians.

b) Find the range of each group.

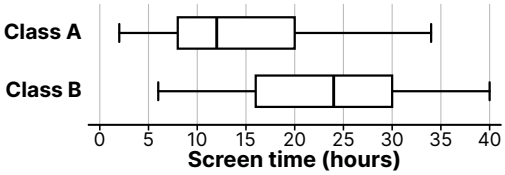
c) Find the interquartile range of each group.

d) In which group is the middle half of the data more spread out?

e) True or false: more than half of the Year 10 group are taller than 170 cm.

f) Describe the shape of each group.

10. The box plots show the weekly screen time, in hours, of two classes.



a) Which group has the higher median? Write down both medians.

b) Find the range of each group.

c) Find the interquartile range of each group.

d) In which group is the middle half of the data more spread out?

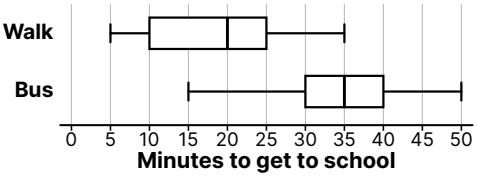
e) True or false: more than half of the Class B group spend more than 20 hours.

f) Describe the shape of each group.

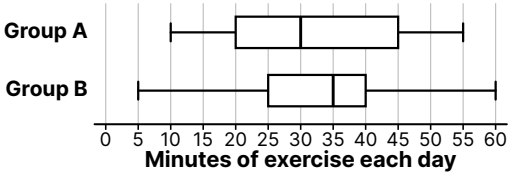
G Making the call ●●●

Remember: You can say one group tends to be higher when the median of one group is outside the box of the other. If each median sits inside the other box, the overlap is too big to make the call.

11. The box plots show how long students who walk and students who take the bus take to get to school. Do students who take the bus tend to take longer? Use the medians and the boxes to explain.



12. The box plots show the minutes of exercise each day for two groups of adults. Does Group B tend to exercise for longer? Use the medians and the boxes to explain.

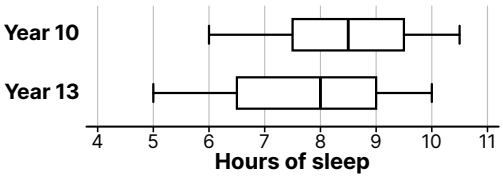


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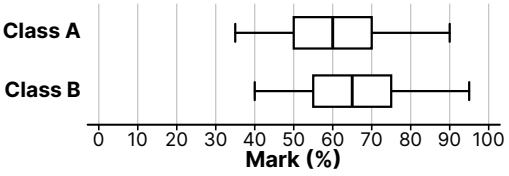
13. The box plots show the prices of second-hand cars at two dealers. Does Dealer 2 tend to charge more? Use the medians and the boxes to explain.



14. The box plots show the hours of sleep reported by Year 10 and Year 13 students. Do Year 10 students tend to sleep for longer? Use the medians and the boxes to explain.



15. The box plots show the marks of two classes in the same test. Does Class B tend to score higher? Use the medians and the boxes to explain.



Extension

16. A set of data has a minimum of 8 and a lower quartile of 15. Its interquartile range is 14, its median is 5 more than its lower quartile, and its range is 36.

a) Find the upper quartile. **b)** Find the median. **c)** Find the maximum.

17. A box plot of 80 values has minimum 10, LQ 18, median 25, UQ 36, maximum 52.

a) How many values are at least 18?

b) How many values are between 18 and 25?

c) How many values are above the upper quartile, 36?

d) How many values are below the median, 25?

e) Can you tell how many values are between 12 and 16? Explain.

f) Two sets of data have the same five-point summary. Must they have the same values? Explain.

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